Rakshith Saligram

rakshith.saligram@gatech.edu 3078 Clairmont Rd, #705, Atlanta 30329 +1 (213)-263-1104
prism.gatech.edu/~rsaligram3/ linkedin.com/rakshith-saligram

Education

2019 – Present

Georgia Institute of Technology, Atlanta, Georgia
PhD, School of Electrical and Computer Engineering
Advisor: Arijit Raychowdhury

2014 – 2016

University of Southern California, Los Angeles, California
M.S. Honors, Ming Hsieh Department of Electrical Engineering
Major: VLSI Design and Computer Engineering

2009 – 2013

BMS College of Engineering, Bangalore, Karnataka
B.E. Electronics and Communication Engineering
Thesis: Design of Fault Tolerant ALU using Parity Preserving Reversible Logic Gates

Academic & Research Experience

2019 – Present

Integrated Circuits and Systems Research Lab
Graduate Research Assistant
Area of Interest: Cryogenic CMOS Circuits, VLSI, High Speed IO Design
My focus is on designing High Performance Computing Systems; I am involved in 2 projects: Cryogenic CMOS VLSI where I am modelling devices in TCAD to explore device characteristics at Cryogenic Temperatures. The TCAD models have been calibrated with experimental verification. The extended results show 10x EDP improvement for simple 32bit ALU. The basic Figure of Merit Circuits including but not limited to DRAM and ALU have been taped out. The second project is study of higher order modulation techniques for chiplet to chiplet interconnects.

Fall 2015

USC Ming Hsieh Institute - Department of Electrical Engineering
Course Development Assistant
Assist the Department Chair of EE Systems to develop and improve EE477L (MOS VLSI Circuit Design) course curriculum.

Summer 2015

USC Ming Hsieh Institute - Department of Electrical Engineering
Course Grader Modern Solid-State Devices
Provide solutions and grade the assignments for EE537- Modern Solid-State Devices course and hold office hours for discussion.

2013 – 2014

Department of ECE, BMS College of Engineering
Lecturer
Instructed freshmen, sophomore & junior classes and senior lab. Visit my homepage for details about the classes. I also functioned as one of the department coordinators for National Board of Accreditation (NBA)

Industry Experience

Summer 2020

ARM Inc.
Intern Research
Worked on benchmarking ARM cores at cryogenic Temperatures, Recharacterization of standard cells at lowered Vdd and iso-performance, self heating analysis

2016-2018

Intel Corporation
Graphics Hardware Engineer in Dual Sub-Slice Marco (DSSM)
Resposible for RTL to GDS-II of multiple generations of Intel's GFx Processors. Worked on timing analysis, timing and routing closure, ECO, Performance verification at block, partition, section, slice and full-chip levels

Research Publications

Journal Articles

1. Bairy, B., Craig, T. S., Gonde, K., Gupta, N., Prajogi, A., Wilner, M., & **Saligram**, **R.** (2016b). Mitigating the impact of NBTI and PBTI Degradation. *Global Journal of Technology and Optimization*, *7*, 195.
2. **Saligram**, **R.**, Hegde, S., Kulkarni, S., Bhagyalakshmi, H., & Venkatesha, M. (2013a). Design of Parity Preserving Logic Based Fault Tolerant Reversible Arithmetic Logic Unit. *International Journal of VLSI Design and Communication Systems*, *10*.
3. **Saligram**, **R.**, Hegde, S., Kulkarni, S., Bhagyalakshmi, H., & Venkatesha, M. (2013b). Design of Fault Tolerant Reversible Multiplexer Based Multi-Boolean Function Generator using Parity Preserving Gates. *International Journal of Computer Applications*.
4. **Saligram**, **R.**, & Ravishankar, R. (2012a). Design of Reversible Multipliers for Linear Filtering Applications in DSP. *International Journal of VLSI Design and Communication Systems*.
5. **Saligram**, **R.**, & Ravishankar, R. (2012b). Novel Code Converter Employing Reversible Lgic. *International Journal of Computer Applications*.

Conference Proceedings

1. **Saligram**, **R.**, Cao, N., Chakraborty, W., Cao, Y., Datta, S., & Raychowdhury, A. (2021). Power-Performance Analysis of Digital Cells for HKMG CMOS 28nm Process Node at Cryogenic Temperatures, In *79th IEEE Device Research Conference (DRC) 2021*, Virtual, USA.
2. Chakraborty, W., Aabrar, K. A., Gomez, J., **Saligram**, **R.**, Raychowdhury, A., & Datta, S. (2021). Cryogenic RF CMOS on 22nm FDSOI Platform with Record ft=495GHz and fMAX=497GHz, In *VLSI Technology Symposia, 2021*, Virtual, Japan.
3. **Saligram**, **R.**, Datta, S., & Raychowdhury, A. (2021). Cryomem: A 4K-300K 1.3GHz eDRAM Macro with Hybrid 2T-Gain-Cell in a 28nm Logic Process for Cryogenic Applications, In *IEEE Custom Integrated Circuits Conference (CICC) 2021*, Virtual, USA.
4. **Saligram**, **R.**, Prasad, D., Pietromonaco, D., Raychowdhury, A., & Cline, B. (2021). A 64-Bit Arm CPU at Cryogenic temperatures: Design Technology Co-Optimization for Power and Performance, In *IEEE Custom Integrated Circuits Conference (CICC) 2021*, Virtual, USA.
5. **Saligram**, **R.**, Kaul, A., Bakir, M., & Raychowdhury, A. (2020). A Model Study of Multilevel Signalling for High-Speed Chiplet-to-Chiplet Communication in 2.5D Integration, In *Proceedings of 28th IFIP/IEEE International Conference on Very Large Scale Integration (VLSI-SoC 2020)*, Salt Lake City, USA.

6. **Saligram**, **R.**, Abhilash, P., & Patel, K. (2018). Realization of Multivalued Logic Combinational Circuits in Fully Depleted Silicon on Insulator, In *Proceedings of IEEE International Conference on Networking, Embedded and Wireless Systems*, Bangalore.
7. **Saligram**, **R.**, Jyoti, K., & Patel, K. (2018). Quarternary digital circuits design using carbon nanotube fets, In *Proceedings of IEEE International Conference on Networking, Embedded and Wireless Systems*, Bangalore.
8. Bairy, B., Craig, T. S., Gonde, K., Gupta, N., Prajogi, A., Wilner, M., & **Saligram**, **R.** (2016a). Towards mitigating the impact of NBTI and PBTI Degradation, In *Proceedings of 2nd World Congress on Automation and Robotics Conference*, Pennsylvania.
9. **Saligram**, **R.** (2013a). Design and Implementation of Logical Cost Efficient Nanometric Fault Tolerant Reversible BCD Adder, In *Proceedings of IEEE 10th INDICON*, IIT Bombay.
10. **Saligram**, **R.** (2013b). Design of Low Logical Cost Conservative Reversible Adders using Novel PCTG, In *Proceedings of IEEE 4th International Symposium on Electronic System Design*, Singapore.
11. **Saligram**, **R.**, & Ravishankar, R. (2013a). Design of Low Logical Cost Adders using Novel Parity Conserving Toffoli Gate, In *Proceedings of IEEE International Conference on Emerging Trends in Communication, Control, Signal Processing and Computing Applications*.
12. Ravishankar, R., & **Saligram**, **R.** (2013a). Design of high speed low power multiplier using reversible logic: A vedic mathematical approach, In *Proceedings of IEEE International Conference on Circuits Power and Computing Technologies*.
13. Ravishankar, R., & **Saligram**, **R.** (2013b). Parity preserving logic based fault tolerant reversible ALU, In *Proceedings of IEEE International Conference on Information and Communication Technologies*.
14. **Saligram**, **R.**, & Ravishankar, R. (2013b). Contemplation of synchronous gray code counter and its variants using reversible logic gates, In *Proceedings of IEEE International Conference on Information and Communication Technologies*.
15. **Saligram**, **R.**, & Ravishankar, R. (2013c). Optimized reversible vedic multipliers for high speed low power operations, In *Proceedings of IEEE International Conference on Information and Communication Technologies*.
16. **Saligram**, **R.**, & Ravishankar, R. (2013d). Towards the Design of Fault Tolerant Reversible Circuits Components of ALU using New PCMF Gate, In *Proceedings of IEEE International Conference on Advances in Computing, Communication and Informatics*.

Books and Chapters

1. **Saligram**, **R.**, Kaul, A., Bakir, M. S., & Raychowdhury, A. (2021). Multilevel Signalling for High-Speed Chiplet-to-Chiplet Communication, In *VLSI-SoC New Technology Enabler*. Springer.
2. Chen, Z., Gonde, K., Ravicz, K., **Saligram**, **R.**, Schlesinger, M., & Wilner, M. (2016). Concluding remarks, In *Wireless Computing in Medicine: From Nano to Cloud with Ethical and Legal Implications*. Wiley.

Tools

EDA	Synopsys Design Compiler, Prime Time, IC-Compiler (ICC/ICC2), Lynx, Cadence NC Sim, Genus, Innovus, Encounter, Virtuoso, Mentor Graphics Modelsim, Calibre
Modelling	Synopsys TCAD Sentaurus, Ansys HFSS, Keysight ADS.
Scripting	Tcl, Perl
HDL	Verilog, VDHL
OS	Windows, RHEL, Ubuntu

Academic Projects

1. DDR3 Memory Controller
2. Modelling of Thin Film Silicon on Glass Enhancement P-MOSFET in HSPICE
3. Performance analysis of out-of-order superscalar processor
4. Implementation of Ternary CAM for Forwarding Modles of Routing Tables in IP using Perl
5. Tomasulo CPU with dynamic instruction scheduling, OoO execution and in order retirement
6. Implementation of FinFETs Logic Gates and comparison of PPA of IG, LP, SG, IG/LP Modes
7. Full Custom Design of 5 stage pipeline in Cadence
8. Design of Asynchronous Network on Chip with Forward Error Correction
9. Implementation and Performance Analysis of DRAM Cell using HSPICE, Perl and MATLAB
10. Design of 6 bit 2's complement Multiplier
11. Design and performance characterization of CMOS Op-Amp and Design of precision FWR
12. 512 Bit SRAM Design
13. Design of CMOS Fibonacci Sequence Generator
14. RTL Design of single clock and two clock FIFO for CDC
15. PODEM Simulator
16. RTL Design of Maze Layout Router using Lee's Algorithm

Awards and Achievements

Academic

2020 **GT-ORNL PhD Seed Research Grant Award**, *Cryogenic CMOS interface circuits for superconducting quantum circuits.*

2016 **MS Honors**, *USC Ming Hsieh Institute-Department of Electrical Engineering.*

2012 **Best paper award at IEEE International Conference on Circuits Power and Computing Technologies**, *Design of High Speed Low Power Multiplier using Reversible logic: a Vedic Mathematical Approach*

2009 **Gnana Shree State Award & Vidya Ratna Puraskar**, *10th Rank in Karnataka State Pre-University, Bangalore, India*

Corporate

2018 **Team Recognition**, *by Section Execution Owner for Pioneering partition execution to foster clear & realistic utilization targets.*

Team Recognition, *by Section Execution Owner for achieving LV closure on two partitions through contingency support.*

2017 **Team Recognition**, *by Section Execution Owner for Pioneering partition LV release for DSSM Section Aostepping.*

Awards and Achievements (continued)

- **SQO Recognition**, *by Caliber COE for discovering Innovative methods for fixing Max-Cap Violations on RP-cells with no degradation in other metrics.*
- **SQO Recognition**, *by Full Chip Caliber Owner for flagging and Pioneering Caliber Issues at FC level and DSSM Section Level.*
- **Team Recognition**, *by Section Execution Owner for supporting multiple section PV to clean up timing violations due to standard cell library changes and for Supporting multiple iterations of Black-box FC-ILM PV to converge inter section paths.*
- **Team Recognition**, *by Section Execution Owner for hacking central SD RTL models to match SSM Floorplan hierarchy and releasing central synthesis netlist, section Bitblast netlist for Floor planning*
- **Team Recognition**, *by Section Execution Owner appreciating efforts in flagging and resolving routing issues with 10nm++ additional metal stacks in ICC2 with Gen11 and Gen12 RTL*

2016
- **Team Recognition**, *by Section Execution Owner for efforts in launching multiple section PV and Caliber*

Talks

Invited

2020
- **Cryogenic CMOS VLSI Circuits**, *ARM Inc.*

2019
- **Modelling of High-Speed Channels**, *Liaison Meeting at Semiconductor Research Corporation*

Academic Activities

Reviewer

2017
- **IEEE Transactions on VLSI**, September

2016
- **Microelectronics Journal Elsevier**, November
- **Alexandria Engineering Journal**, July

2015
- **ACM Journal on Emerging Technology in Computing**, August
- **Microelectronics Journal Elsevier**, July
- **Microelectronics Journal Elsevier**, January

Organization

2014
- **Digital Front End Design and Implementation using Synopsys Tools**, Two Week Faculty Development Program
- **Antenna Technologies**, One week Faculty Development Workshop